OLA Ride Booking Analysis Project

Tool Used:- Power BI | SQL | Excel | DAX

1. Problem Statement & Business Questions

OLA was facing a high number of ride cancellations, affecting customer experience and operational efficiency.

This project was created to analyze:
- Why rides were getting cancelled
- Which vehicle types generated the most revenue
- Customer booking behavior
- Revenue trends
- Service quality based on ratings

2. Dataset Overview

This project was created to analyze OLA ride booking operations and understand booking trends, cancellations, customer behavior, revenue performance, and service quality using **SQL and Power BI.**

The dataset contained around **30,000 ride booking** records for a **one-month** period. Data cleaning was performed in Excel before loading the dataset into Power BI for dashboard creation and KPI analysis.

3. Why Excel Was Used as Primary Data Source

Excel was used as the primary data source because the dataset was available in raw spreadsheet format and required cleaning before analysis.

Data Cleaning steps:

- Removed formatting inconsistencies
- Checked null values
- Standardized booking status values
- Cleaned text columns
- Validated booking value calculations

After cleaning the dataset was imported into **Power BI for analysis and dashboard creation**

4. Why SQL Was Used in This Project

SQL was used to analyze **booking data & solve business-related problems before visualization.**

It helped in:

Filtering required data, Validating KPI calculations, Identifying booking patterns, Analyzing cancellations, & Rretrieving business insights efficiently.

The project was developed **using MySQL Workbench** with one **main table: bookings**.

5. Main KPIs Used in Dashboard

1. Total Bookings
2. Average Rating
3. Total Booking Value
4. Vehicle-wise Booking Value
5. Total Successful Booking Value
6. Average Ride Distance
7. Total Ride Distance
8. Top 5 Customers
9. Driver Cancellation Rate
10. Customer Cancellation Rate
11. Overall Cancellation Rate

Why Particular KPIs Were Selected

Why Total Booking Value KPI Was Used

This KPI was used to measure overall business revenue generated from ride bookings.

Why Cancellation Rate KPI Was Important

Cancellation rate helped identify operational inefficiencies and ride completion issues.

Why Average Rating KPI Was Used

Ratings were used to understand customer satisfaction and service quality.

Why Top 5 Customers Analysis Was Performed

This helped identify repeat customers and high-engagement users who contribute significantly to bookings.

SQL Queries Used for Analysis

Retrieve Successful Bookings:

create view successful_bookings AS

```
SELECT * FROM bookings

WHERE TRIM(`Booking Status`) = 'Success';

SELECT * FROM successful_bookings;
```

Top 5 Customers by Bookings:

```
CREATE VIEW top_5_customers_who_booked_the_highest_number_of_ride AS

select trim(`Customer ID`), COUNT(`Booking ID`) as Total_Rides

from bookings

group by TRIM(`Customer ID`)

ORDER BY Total_Rides desc

LIMIT 5;

SELECT  * FROM top_5_customers_who_booked_the_highest_number_of_ride;
```

Why This Query Was Used

To identify repeat customers and understand customer retention behavior.

Result

Some customers showed frequent ride bookings, indicating strong repeat usage patterns.

Maximum and Minimum Driver Ratings for Prime Sedan Bookings:-

Used this query was used to analyze the service quality of the Prime Sedan vehicle category by finding the highest and lowest driver ratings received for completed rides.

```
REATE VIEW max_and_min_driver_rating_for_prime_sedan AS

SELECT

  MAX(CAST(`Driver Ratings` AS DECIMAL(3,1))) AS maximum_ratings,

  MIN(CAST(`Driver Ratings` AS DECIMAL(3,1))) AS minimum_ratings

FROM bookings

WHERE TRIM(`Vehicle Type`) = "Prime Sedan"

AND `Driver Ratings` IS NOT NULL

AND `Driver Ratings` <> '';

SELECT * FROM max_and_min_driver_rating_for_prime_sedan;
```

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6. DAX Measures Used

Overall Cancellation Rate:- **Used to measure total ride cancellations compared to total bookings.**

```
Overall Cancellation Rate =
DIVIDE(
  CALCULATE(
    COUNTROWS(bookings),
    bookings[Booking Status] = "Cancelled by Driver"
  )
  +
  CALCULATE(
    COUNTROWS(bookings),
    bookings[Booking Status] = "Cancelled by Customer"
  ),
  COUNTROWS(bookings)
)
```

Driver Cancellation Rate:- **Used to identify operational issues from the driver side.**

```
Driver Cancellation Rate =
DIVIDE(
  CALCULATE(
    COUNTROWS(bookings),
    bookings[Booking Status] = "Cancelled by Driver"
  ),
  COUNTROWS(bookings)
)
```

Customer Cancellation Rate:- **Used to analyze customer-side cancellation behavior separately.**

```
Customer Cancellation Rate =
DIVIDE(
   CALCULATE(
      COUNTROWS(bookings),
      bookings[Booking Status] = "Cancelled by Customer"
   ),
   COUNTROWS(bookings)
)
```

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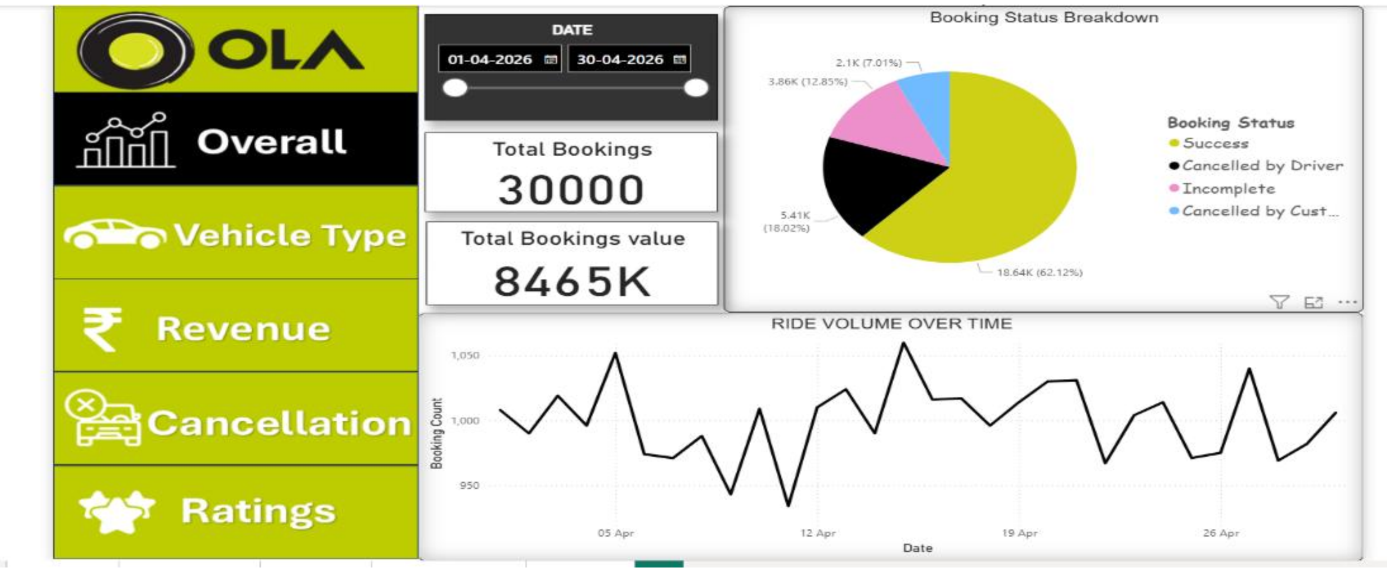
7. Dashboard Understanding

Overall Dashboard

This dashboard provides a high-level overview of overall ride booking operations and business performance.

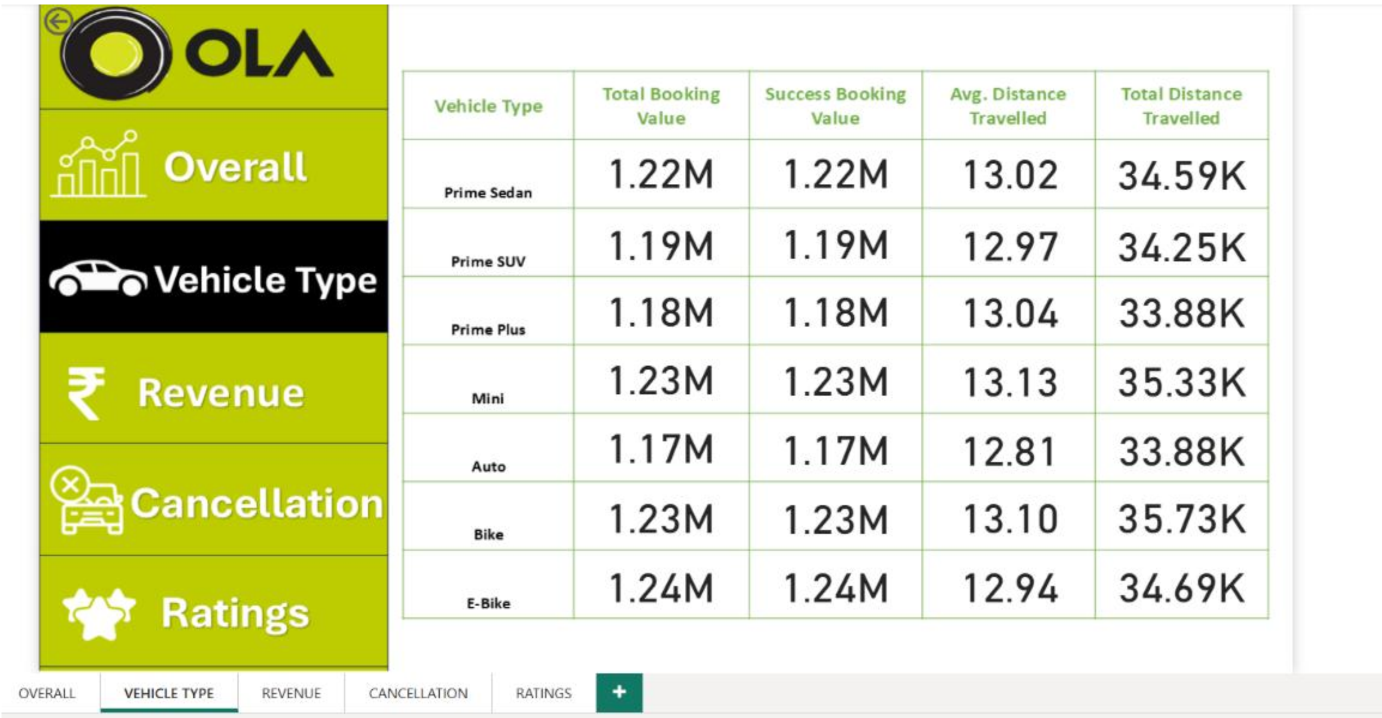
It was used to monitor:-
Total bookings, Booking value, Booking trends over time, & Booking status distribution.



The dashboard helped identify that a significant portion of rides were getting cancelled or marked incomplete, affecting operational efficiency.

Vehicle Type Dashboard

This dashboard was created to compare the performance of different vehicle categories based on ride distance, booking value, and ride demand.



Vehicle Type	Total Booking Value	Success Booking Value	Avg. Distance Travelled	Total Distance Travelled
Prime Sedan	1.22M	1.22M	13.02	34.59K
Prime SUV	1.19M	1.19M	12.97	34.25K
Prime Plus	1.18M	1.18M	13.04	33.88K
Mini	1.23M	1.23M	13.13	35.33K
Auto	1.17M	1.17M	12.81	33.88K
Bike	1.23M	1.23M	13.10	35.73K
E-Bike	1.24M	1.24M	12.94	34.69K

The analysis showed that **E-Bike, Bike, and Mini** were among the **highest-performing vehicle** categories in terms of revenue and ride activity.

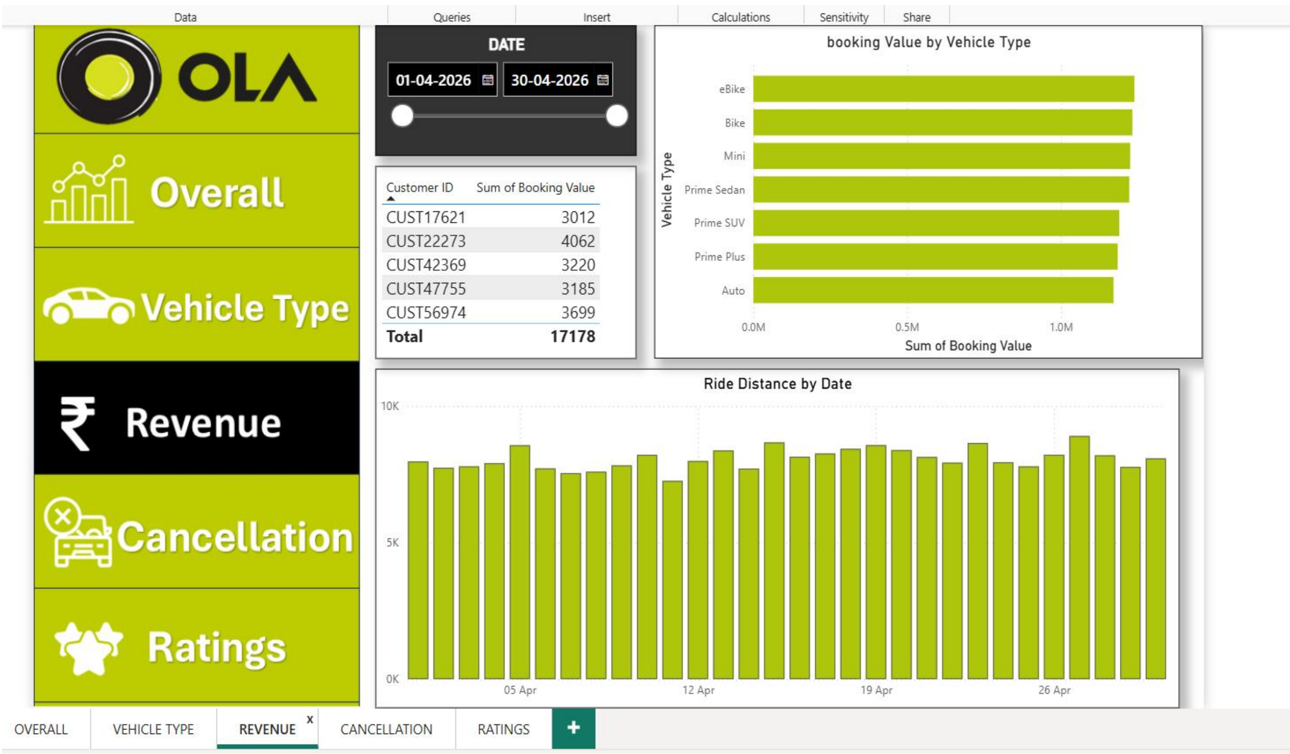
The dashboard helped identify which vehicle types contributed most to business performance.

Revenue Dashboard

Used to analyze:

Booking value, Top customers, Ride distance by date, & Vehicle-wise revenue contribution.

The analysis identified E-Bike, Bike, and Mini as the top revenue-generating vehicle categories



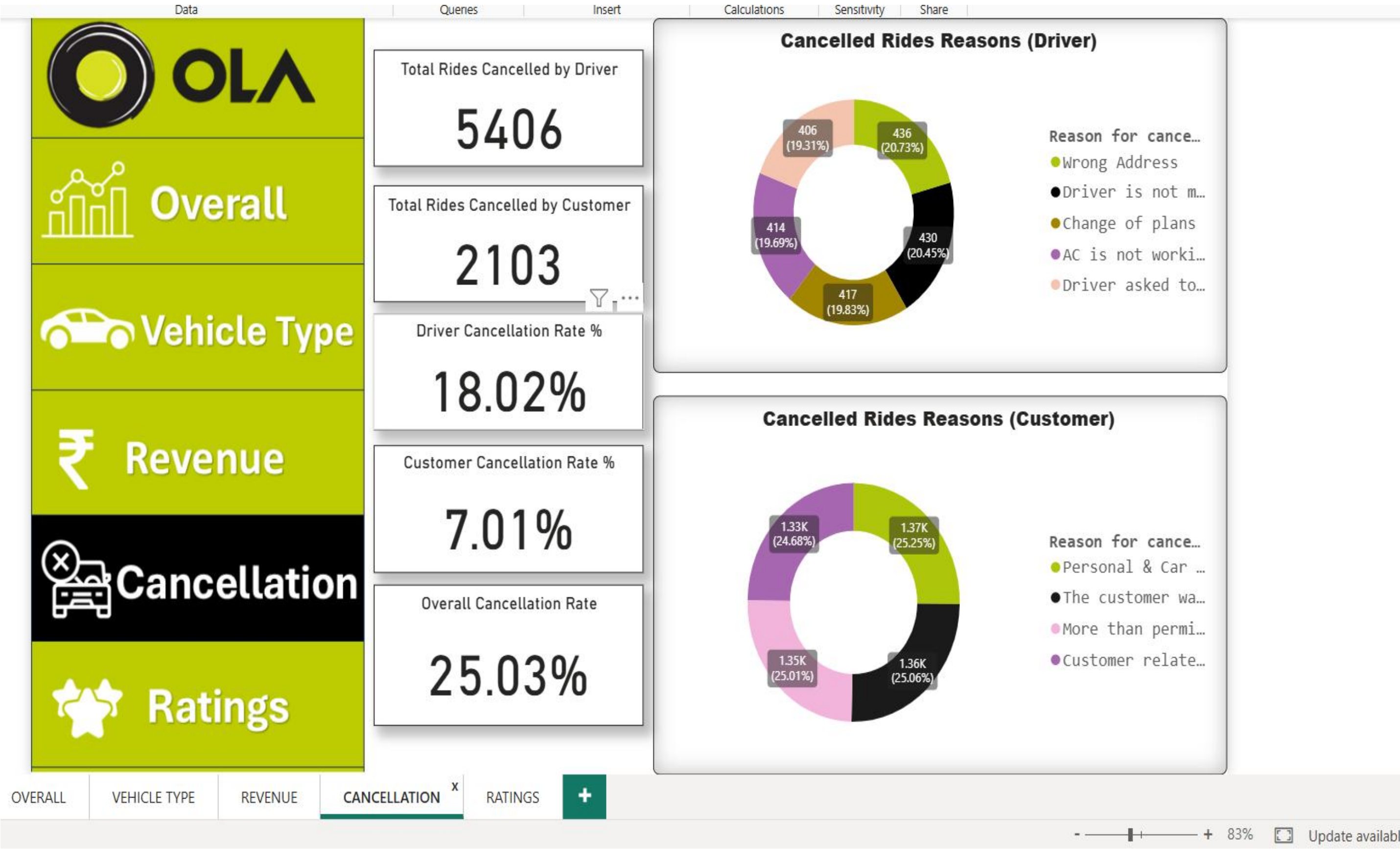
It also helped identify repeat customers with high booking frequency and understand customer spending behavior.

Cancellation Dashboard

Used to understand:

- cancellation reasons,
- driver vs customer cancellations,

operational issues affecting ride completion.



The dashboard showed that driver cancellations (18.02%) were significantly higher than customer cancellations (7.01%), contributing heavily to the overall cancellation rate of 25.03%.

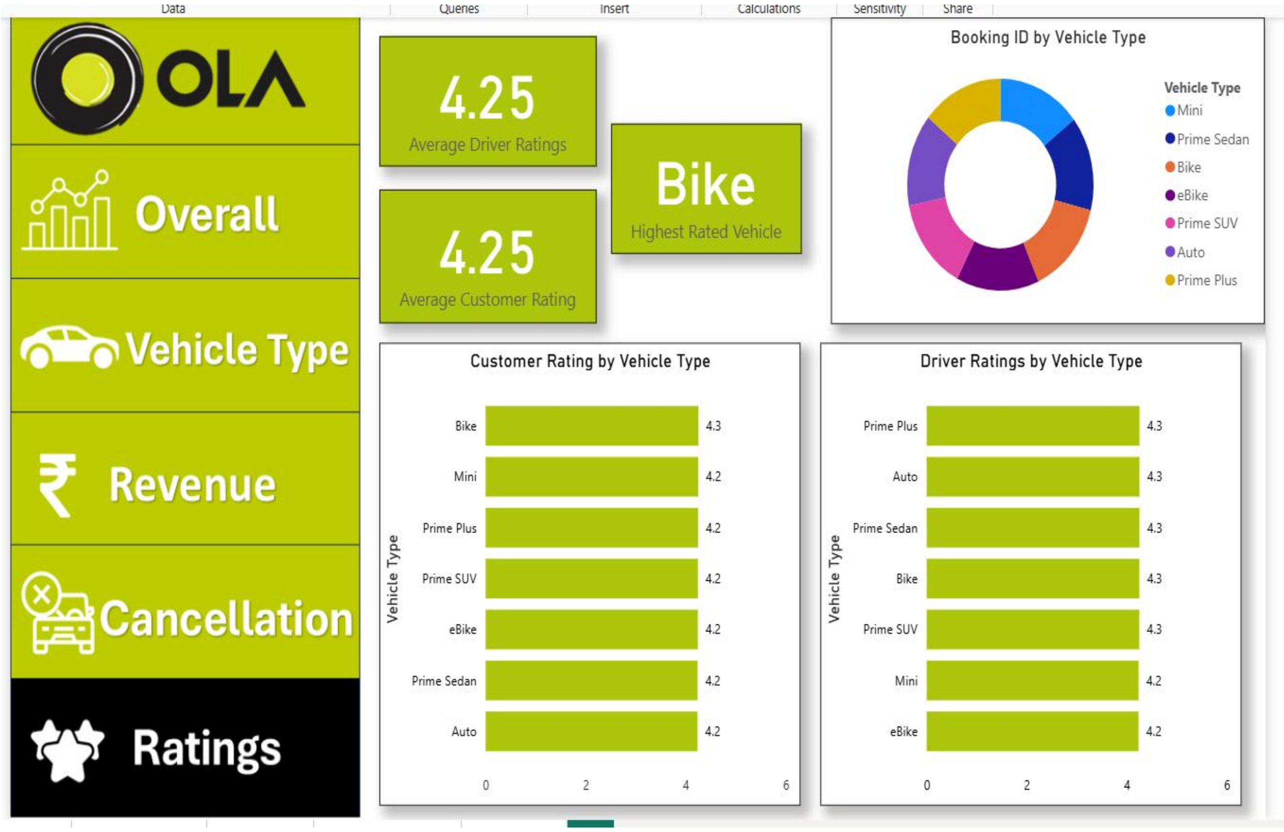
Ratings Dashboard

Used to compare:

- Customer ratings,

- Driver ratings,

- Service quality across vehicle categories.



Average ratings remained mostly between 4.0–4.4.

The dashboard also helped compare vehicle-wise rating performance and identify service quality trend

8. Charts & Visuals Used

Pie Chart

Used for booking status breakdown because it gives a quick comparison between successful, cancelled, and incomplete rides.

Line Chart

Used for ride volume over time to identify daily booking trends and ride fluctuations.

Bar Charts

Used for vehicle type comparison and cancellation reasons because category comparisons are easier to analyze visually.

KPI Cards

Used to highlight important business metrics instantly without applying multiple filters.

9. Key Insights Found

Overall cancellation rate was **25.03%**, meaning approximately **7,500 rides** out of 30,000 bookings were not completed successfully.
2. Driver cancellations were **18.02%**, which was significantly higher than customer cancellations at **7.01%**.
3. E-Bike, Bike, and Mini were the top revenue-generating vehicle categories in the dataset.
4. Average customer and driver ratings remained between **4.0–4.4**, indicating generally positive service quality.
5. The dataset identified repeat booking behavior among top customers, showing strong customer retention activity.
6. Ride booking trends fluctuated across different dates, indicating changing daily ride demand patterns

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10. Challenges Faced During the Project

- Some KPI cards initially showed incorrect totals because of slicer interactions.
- Cancellation calculations first included incomplete rides, resulting in incorrect %
- Booking value totals were affected due to filter context issues in Power BI.

How These Issues Were Solved

- Fixed visual interaction and filter context problems.
- Created separate **DAX** measures for customer and driver cancellations.
- **Used SQL queries** to validate dashboard calculations.
- Rechecked aggregation logic for booking value calculations.

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11. Recommendations

- Driver cancellation rate was **18.02%**, much higher than customer cancellations (**7.01%**). Improving driver monitoring and ride acceptance tracking could help reduce the overall cancellation rate of **25.03%**.
- Around **7,500 rides** were cancelled or incomplete. Reducing customer waiting time and improving ride allocation can help improve ride completion efficiency.
- Common cancellation reasons included **wrong pickup/drop locations. Improving GPS accuracy and address verification** could reduce failed ride attempts.
- **E-Bike, Bike, and Mini generated the highest booking value in the dataset**. Increasing the availability of these vehicle categories could help improve overall revenue performance.
- Top customer analysis showed repeated ride bookings from some users. Introducing loyalty rewards or offers for frequent customers could improve customer retention and repeat bookings.

12. Conclusion

This project helped in understanding **how SQL and Power BI** can be used together to **analyze business operations** and solve real-world problems using data. The dashboards provided useful insights into ride performance, cancellations, customer behavior, and operational efficiency.